

LESSON 4.6

EQUIVALENT EXPRESSIONS



LESSON INTRODUCTION

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

LEARNING TARGETS

- I can determine equivalency using the laws of exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational exponents. Apply laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you determine if expressions are equivalent using the laws of exponents if the expressions do not have the same base?
- How do you use the laws of exponents to rewrite numerical expressions to have a common base?

LESSON NARRATIVE

Students explore how rewriting exponential expressions can be equivalent if they have different bases. Students use the laws of exponents to rewrite exponential expressions to determine if expressions are equivalent.

COMPONENT SUMMARY

4.6.1 WARM-UP (~5 MIN)

Evaluate expressions with integer exponents

4.6.3 GUIDED INSTRUCTION (15 MIN)

Rewrite expressions with different bases

4.6.5 COLLABORATION (10-15 MIN)

Rewrite expressions with as few bases as possible

4.6.2 EXPLORATION (5-10 MIN)

Explore expressions with exponents that have different bases to discover equivalence

4.6.4 GUIDED PRACTICE (10 MIN)

Rewrite expressions with different bases using laws of exponents and assess equivalence

4.6.6 WRAP-UP (~5 MIN)

Reflect on effort made and understanding of the lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Desmos scientific calculator

ADDITIONAL PREPARATION

- Consider strategically grouping students for 4.6.5 Collaboration based on observational data from the previous lessons.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize exponents with different bases that can be rewritten with the same base.

THINGS TO CONSIDER

- Students will need calculators for 4.6.2 Exploration. They do not need calculators for the following activities.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

As students complete 4.6.5 Collaboration, collect and display student work. When displaying, organize the student work by each expression. Engage students in a discussion about what they notice in the strategies used for rewriting each of the same expressions. Look for opportunities to have students recognize when students took different steps to rewrite expressions and discuss if the resulting expressions were equivalent or if they were equal. Ask students to share out if they feel they used a unique strategy.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

4.6.3 Guided Instruction asks students to reflect and discuss how expressions with different bases were rewritten and simplified using the laws of exponents. Prompt students who used different steps to share and discuss similarities and differences between their strategies.

DIFFERENTIATE INSTRUCTION

Provide students the option to verbally explain if they do or do not agree with Xavier in 4.6.4 Guided Practice.

Use observational data from the 4.6.2 Exploration, 4.6.3 Guided Instruction, and 4.6.4 Guided Practice to determine strategic grouping for the 4.6.5 Collaboration. If necessary, pull a small group of students to provide remediation or additional support during the 4.6.5 Collaboration.

Additional differentiation opportunities are denoted throughout the lesson guide.

Expression A	Expression B
$3^2 \cdot 6^3$ $= 3^2 \cdot (3 \cdot 2)^3$ $= 3^2 \cdot 3^3 \cdot 2^3$ $= 3^5 \cdot 2^3$	$\frac{(3 \cdot 2)^5}{2^2}$ $= \frac{3^5 \cdot 2^5}{2^2}$ $= 3^5 \cdot 2^3$

4.6.1 WARM-UP: BELL WORK

TEACHER MOVES

Use student understanding to inform your instruction during this and further lessons. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.



4.6.1 Warm-Up: Bell Work

1. Evaluate each expression. Show your work.

$$\begin{aligned} \text{a. } & \frac{(-4)^2(-4)^{-3}}{-4} \\ &= \frac{(-4)^{-1}}{-4} \\ &= \frac{1}{-4 \cdot -4} \\ &= \frac{1}{16} \end{aligned}$$

$$\begin{aligned} \text{b. } & \left(\left(\frac{2}{5}\right)^3\right)^{-2} \cdot \left(\frac{5}{2}\right)^{-4} \\ &= \left(\frac{2}{5}\right)^{-6} \cdot \left(\frac{5}{2}\right)^{-4} \\ &= \left(\frac{2}{5}\right)^{-6} \cdot \left(\frac{2}{5}\right)^4 \\ &= \left(\frac{2}{5}\right)^{-2} \\ &= \left(\frac{5}{2}\right)^2 \\ &= \frac{25}{4} \end{aligned}$$



4.6.2 Exploration: Exponential Expressions with Different Bases

1. Three expressions are given in the table.
- a. For each given expression, generate an equivalent expression by applying exponent laws. Then, find the value of the expression using a calculator.

Given Expression	Equivalent Expression	Value
$\frac{2^{14}}{2^4}$	2^{10}	1,024
$4^7 \cdot 4^{-2}$	4^5	1,024
$16 \cdot 4 \cdot 4^2$	$16 \cdot 4 \cdot 16$	1,024

- b. Discuss your results with your partner. What do you notice about the three expressions?
Answers will vary. All three expressions have the same value.

4.6.2 EXPLORATION: EXPONENTIAL EXPRESSIONS WITH DIFFERENT BASES

TEACHER MOVES

Demonstrate and explain how to enter exponential expressions in a calculator. Consider having all students use the same calculator, such as the Desmos scientific calculator.

TEACHER MOVES

Students may make errors during the 4.6.2 Exploration when finding an equivalent expression. For example: students may put 8 as the equivalence of 42. Allow the opportunity for them to correct their own thinking as they work through with their partner. Do not be afraid to let the mistakes happen; discuss them during the transition to the next component.

4.6.3 GUIDED INSTRUCTION: REWRITING EXPRESSIONS WITH DIFFERENT BASES

QUESTIONING

After students have completed question 1, see if students are able to rewrite 343 as an equivalent expression with an exponent. If students are ready, ask them to rewrite 256 two different ways as an exponential expression. Responses might be 2^8 and 16^2 and 4^4 .



4.6.3 Guided Instruction: Rewriting Expressions with Different Bases

Exponent laws can be applied to exponents with the **same base**. However, sometimes a base can be rewritten to create a common base when perfect squares, cubes, fourths, etc. are present.

1. Complete the table. The first row is started for you.

Base	Square	Cube	Fourth
2	$2^2 = 4$	$2^3 = 8$	$2^4 = 16$
3	$3^2 = 9$	$3^3 = 27$	$3^4 = 81$
4	$4^2 = 16$	$4^3 = 64$	$4^4 = 256$
5	$5^2 = 25$	$5^3 = 125$	$5^4 = 625$

2. Write an exponential expression equivalent to 8^5 that has a base of 2.

Step	Description
8^5	Determine an expression with base 2 that is equivalent to 8.
$= (2^3)^5$	Substitute the power of 2 for 8 in the expression.
$= 2^{15}$	Apply power of a power law of exponents.

3. Write an exponential expression equivalent to 9^4 that has a base of 3.

$$\begin{aligned}
 &9^4 \\
 &= (3^2)^4 \\
 &= 3^8
 \end{aligned}$$

TEACHER MOVES

Consider having students list the exponent laws used during the process to write equivalent expressions for Questions 3 and 4.

4. Complete the steps to write an exponential expression equivalent to $2^3 \cdot 4^3$ that has a base of 2.

$$\begin{aligned}
 &2^3 \cdot 4^3 \\
 &= 2^3 \cdot (2^2)^3 \\
 &= 2^3 \cdot 2^6 \\
 &= 2^9
 \end{aligned}$$

5. Work with your partner to rewrite each expression from the Exploration using a common base of 2.

Expression A	Expression B	Expression C
$\frac{2^{14}}{2^4}$	$4^7 \cdot 4^{-2}$	$16 \cdot 4 \cdot 4^2$
2^{10}	$(2^2)^7 \cdot (2^2)^{-2}$	$2^4 \cdot (2^2) \cdot (2^2)^2$
	$2^{14} \cdot (2^{-4})$	$2^4 \cdot (2^2) \cdot 2^4$
	2^{10}	2^{10}



4.6.4 Guided Practice: Different Bases, Same Value

The laws of exponents can be applied to determine if expressions are equivalent.

1. Consider the expressions $3^2 \cdot 6^3$ and $\frac{(3 \cdot 2)^5}{2^2}$.
 - a. Complete the steps to rewrite each expression using laws of exponents.

Expression A	Expression B
$3^2 \cdot 6^3$	$\frac{(3 \cdot 2)^5}{2^2}$
$= 3^2 \cdot (\boxed{3} \cdot 2)^3$	$= \frac{\boxed{5} \cdot \boxed{5}}{2^2}$
$= 3^2 \cdot \boxed{3} \cdot \boxed{2} \cdot \boxed{3}$	$= 3^{\boxed{5}} \cdot 2^{\boxed{3}}$
$= 3^{\boxed{5}} \cdot 2^{\boxed{3}}$	

- b. Complete the statement.

The expressions $3^2 \cdot 6^3$ and $\frac{(3 \cdot 2)^5}{2^2}$ are

☒ equivalent
☐ not equivalent because they simplify to

☒ the same expression.
☐ different expressions.

2. Xavier says the expressions $(\frac{1}{4})^2 \cdot 2^{-3}$ and $(\frac{1}{2})^4 \cdot 4^{-1}$ are equivalent because both expressions can be rewritten as 2^{-7} .

- a. Generate an equivalent expression for each expression using a base of 2. Show your work.

$(\frac{1}{4})^2 \cdot 2^{-3}$	$(\frac{1}{2})^4 \cdot 4^{-1}$
$4^{-2} \cdot 2^{-3}$	$2^{-4} \cdot 4^{-1}$
$(2^2)^{-2} \cdot 2^{-3}$	$2^{-4} \cdot (2^2)^{-1}$
$2^{-4} \cdot 2^{-3}$	$2^{-4} \cdot 2^{-2}$
2^{-7}	2^{-6}

- b. Explain whether or not you agree with Xavier.

I disagree with Xavier because the two expressions will not both simplify to 2^{-7} . The second expression simplifies to 2^{-6} not 2^{-7} .

4.6.4 GUIDED PRACTICE: DIFFERENT BASES, SAME VALUE

DIFFERENTIATE INSTRUCTION

Allow informal observation throughout the first three activities to dictate how “guided” this 4.6.4 Guided Practice should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

QUESTIONING

If students are ready for a challenge, ask them to determine whether the following expressions are equivalent.

$(\frac{6^2 \cdot 10^{-7}}{30^{-7}})$ and $3^7 \cdot (3 \cdot 2)^2$

4.6.5 COLLABORATION: EQUIVALENT EXPRESSIONS

TEACHER MOVES

Collect and Display

Collect and display student work. Organize by each expression. Then engage students in a discussion about which laws of exponents they used and what they notice about the expressions when different laws were used.

TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the 4.6.6 Wrap-Up. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.

ENRICHMENT

Do you agree with the steps your partner took? Is there an alternative sequence of steps possible to generate the same equivalent expression? How do you know that your expression includes the fewest number of bases possible?



4.6.5 Collaboration: Equivalent Expressions

1. With your partner, determine who is Partner A and who is Partner B.
 - a. Generate an equivalent exponential expression for each expression in your column using as few bases as possible.

Partner A		Partner B
$3^3 \cdot 3^4$	3^7	$\frac{3^9}{3^2}$
$\frac{2^{-2}}{2^{-4}} \cdot 7$	$2^2 \cdot 7$	$\frac{(7 \cdot 2)^2}{7}$
$2^2 \cdot 4^2$	4^3 or 2^6	$4^5 \cdot 4^{-2}$
$\frac{3^5}{6^2}$	$\frac{3^3}{2^2}$	$3 \cdot \left(\frac{2}{3}\right)^{-2}$
$\frac{5^2 \cdot 5^0}{2^{-4} \cdot 2^3}$	$2 \cdot 5^2$	$\frac{10^4}{5^2 \cdot 2^3}$

- b. The expressions in each row are equivalent. If your expression and your partner's expressions are not equivalent, work together to find and correct any error(s). Then, write the equivalent expression in the middle column.



4.6.6 Wrap-Up: Reflection

1. Complete the table as you reflect on today’s lesson.
Answers will vary.

Today I deserve  for ...	
Today I  by doing ...	
Today I was  by ...	

4.6.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider providing students with sentence stems and/or answer choices. Prompt students to look back at their work on each component of the lesson as they compose their responses.

ENRICHMENT

- What do you deserve applause for?
- How did you excel?
- What parts of the lesson were confusing?
- How can you use this reflection to inform your planning, instruction, and assessment practices for future lessons?

DIFFERENTIATE INSTRUCTION

Consider taking note of what students were confused by to create a plan to address any items students are still confused by in a future lesson.